# When Fairness Breaks Down: Counterfactual Analysis of AdaFair on Small Imbalanced Data

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## Abstract

AdaFair is a fairness-aware machine learning algorithm designed to mitigate discriminatory outcomes by incorporating group fairness constraints during training. Although AdaFair has demonstrated strong performance on several benchmark datasets, prior work identified substantial performance degradation on the positive non-protected subpopulation in the Communities and Crime dataset Carree [2025]. This study investigates the factors contributing to this failure using counterfactual explanations generated with DiCE. The analysis reveals that the protected attribute `racepctblack` is the most critical feature within the counterfactuals, and when excluded, the model relies on proxy features strongly correlated with the protected attribute. Most critically, t-SNE visualization demonstrates that instance location within the feature space is the primary determinant of performance: minority instances embedded within majority-class regions achieve substantially lower accuracy. These findings indicate that AdaFair struggles when extreme class imbalance is combined with protected attribute imbalance and unfavorable spatial distribution.

## 1 Introduction

Since the introduction of the European AI Act in 2024, algorithmic systems deployed within the European Union are required to protect public interests regards health, safety and fundamental rights European Parliament and Council of the European Union [2024]. This regulatory development has motivated substantial research into fairness-aware machine learning methods. One such approach is AdaFair, an in-processing algorithm designed to enforce fairness constraints during model training Iosifidis and Ntoutsi [2019]. In its original publication, AdaFair demonstrated strong predictive performance across several benchmark datasets, including KDD Census Income Bureau [2000], Adult Census Income Becker and Kohavi [1996], Bank Marketing Moro et al. [2012], and COMPAS Angwin et al. [2016].

However, during evaluation on Communities and Crime a small and highly imbalanced data set, a substantial decline in performance was observed for a specific subpopulation Carree [2025]. Notably, this subpopulation corresponded to the group with the fewest instances in the dataset. Attempts to mitigate this issue through upsampling did not lead to measurable performance improvements Carree [2025]. Consequently, an open question remains as to why the model fails to effectively classify this subpopulation.

This issue is significant both from a technical and a legal perspective. From a technical point of view, the extreme class imbalance poses a major challenge, particularly when instances belonging to minority subpopulations are highly similar to those of majority groups Brzeziński et al. [2024]. Understanding how fairness-aware models behave under such conditions is relevant, not only for AdaFair but for machine learning models more broadly, as real-world data often exhibit severe imbalances Branco et al. [2016].

From a legal and ethical perspective, the objective of fairness-aware algorithms is to mitigate discriminatory outcomes, including racial bias against protected subpopulations. However, if a model cannot adequately handle proportion imbalance, it may still produce unfair or biased predictions

despite incorporating fairness constraints. Therefore, it is essential to investigate how AdaFair processes highly imbalanced data, identify the factors that contribute to the observed performance degradation for minority subpopulations, and explore potential strategies to prevent this.

## 2 Related Work

AdaFair is a fairness-aware ensemble learning method that extends the AdaBoost algorithm by incorporating group fairness constraints directly into the boosting process Iosifidis and Ntoutsi [2019]. In standard AdaBoost, instance weights are updated exclusively based on classification errors Schapire [1999], which can increase disparities in predictive performance across protected and unprotected groups. AdaFair modifies this process by incorporating a cumulative fairness measure that prioritizes misclassified and unfairly treated instances, addresses class imbalance, and jointly optimizes for both predictive accuracy and fairness Iosifidis and Ntoutsi [2019].

DiCE (Diverse Counterfactual Explanations) proposed by Mothilal et al. [2020] is a model-agnostic framework for generating diverse counterfactual explanations in machine learning models. It identifies minimal feature perturbations that alter a model's prediction to a desired outcome. To ensure diversity of counterfactuals, DiCE uses determinantal point processes (DPPs), which measure the diversity of a set of counterfactuals via the determinant of a kernel matrix $K$, defined as

$$K_{i,j} = \frac{1}{1 + \text{dist}(c_i, c_j)},$$

where $\text{dist}(c_i, c_j)$ is a distance metric between counterfactuals $c_i$ and $c_j$. The diversity score is given by

$$\text{dpp_diversity} = \det(K).$$

Importantly, summaries of these counterfactual sets can be used to estimate feature importance. Features that are changed more frequently across proximal counterfactuals are interpreted as more necessary for the original model output, since altering them is more likely to flip the prediction Mothilal et al. [2020].

## 3 Approach

This project approach is divided into 3 sections: replication of prior results by Carree [2025], generation and evaluation of counterfactuals, evaluation of the positive non-protected instances for AdaFair trained on downsampled and on counterfactual data. Experiments were implemented in Python 3.8 due to AdaFair dependency constraints, leveraging pandas and NumPy for data processing, scikit-learn for model training and evaluation, AdaFair and DiCE for fairness and counterfactual analysis, and matplotlib and seaborn for visualization. The AdaFair models were trained with standard hyperparameter settings using stratified shuffle-split cross-validation to ensure balanced representation across folds, with performance metrics averaged across five runs.

### 3.1 Replication of Prior Results

The dataset was preprocessed following the same approach as in the prior work. Because AdaFair requires class labels encoded as 1 and $-1$, the target variable was binarized as 1 (ViolentCrimesPerPop $\geq 0.7$) to indicate high crime rates and $-1$ otherwise Iosifidis and Ntoutsi [2019]. The sensitive attribute `racepctblack` was binarized such that communities with a Black population $\geq 6\%$ were labeled as protected (1) and those below $6\%$ as non-protected (0), following Le Quy et al. [2022]. Both AdaBoost and AdaFair were trained on this dataset, yielding results consistent with Carree [2025]. The performance metrics were computed on each split and aggregated by reporting the mean and standard deviation across runs. The five resulting models were saved for use with DiCE, along with their predictions on all positive non-protected instances.

### 3.2 Counterfactual Generation and evaluation

First, a DiCE data object was created from the original dataset. Counterfactuals were generated both with and without allowing changes to the protected attribute. Using DiCE's local feature importance

tool, 20 counterfactuals per positive non-protected instance were generated for each of the five models. Feature importance was quantified by averaging the local importance scores across all instances for each model and then across all models. Separately, the magnitude of change per feature was calculated by measuring the distance between each counterfactual and its original instance, again averaging over instances and models. For the five most important features, correlations with the protected attribute were computed. Finally, counterfactuals were visualized alongside the original data using t-SNE.

### 3.3 AdaFair Evaluation Across Datasets

Performance on the positive non-protected instances was also evaluated on a downsampled dataset, where the other subpopulations were reduced to 115 instances, matching the number of positive protected instances. Evaluation was additionally performed on a dataset comprising both the original data and all generated counterfactuals, to assess whether their inclusion could improve performance on the positive non-protected group. Each positive non-protected instance was subsequently assigned a unique identifier (ID) based on a hash of its feature values, excluding the protected attribute, to allow consistent tracking across datasets. Model performance was then compared across the three types of datasets using these IDs, and instances were plotted along with their IDs to examine how their position within the data influenced their performance.

## 4 Experiments

Table 1 displays the performance of both AdaBoost and Adafair on the Communities and Crime dataset. These results correspond to the work of Carree [2025] with a noticeable drop in performance of both models for the positive non-protected group. To find the reason for the performance drop in AdaFair, counterfactual importance was obtained both with and without the ability to change the protected attribute. The results are shown in Figure 1, with the most important feature being the protected attribute when it was allowed to change.

Table 1: Performance comparison (mean ± standard deviation over five runs).

| | AdaBoost (mean ± std) | AdaFair (mean ± std) |
|---|---|---|
| balanced_accuracy | 0.688 ± 0.035 | 0.736 ± 0.048 |
| accuracy | 0.933 ± 0.007 | 0.915 ± 0.014 |
| fairness | 0.492 ± 0.084 | 0.456 ± 0.101 |
| TPR_protected | 0.434 ± 0.077 | 0.554 ± 0.119 |
| TPR_non_protected | 0.0 ± 0.0 | 0.157 ± 0.151 |
| TNR_protected | 0.938 ± 0.014 | 0.911 ± 0.029 |
| TNR_non_protected | 0.996 ± 0.003 | 0.97 ± 0.013 |

The results also show that there is no significant change for the other features when `racepctblack` is not allowed to change. Therefore, for the five most important features, their magnitude of change and their correlation with the protected attribute were calculated, shown in Table 2. The two features requiring the largest average changes to flip predictions, `PctKids2Par` ($-0.291$) and `PctIlleg` ($0.204$), also exhibit strong correlations with the protected attribute `racepctblack` ($r = -0.498$ and $r = 0.496$, respectively), suggesting the model may be relying on race-proxy features. However, the large magnitude of the required changes may instead reflect the difficulty of the underlying data distribution, making it difficult to distinguish algorithmic bias from inherent distributional challenges.

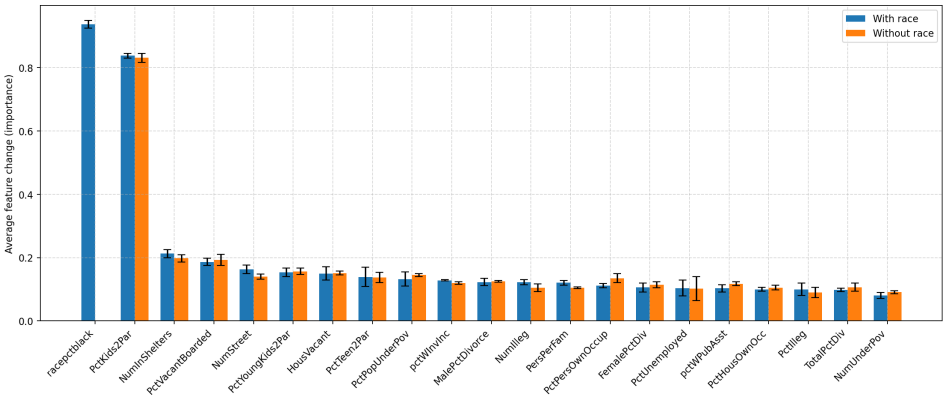


Figure 1: DiCE feature importance for the top features, comparing models with and without the protected attribute `racepctblack`. Error bars indicate the standard deviation across seeds.

Table 2: Top 5 features by average counterfactual change and their correlation with the protected attribute racepctblack.

| | Feature Change (avg) | Correlation with racepctblack |
|---|---|---|
| PctKids2Par | -0.291 | -0.498 |
| PctIlleg | 0.204 | 0.496 |
| HousVacant | 0.166 | 0.247 |
| NumStreet | 0.136 | 0.142 |
| NumInShelters | 0.106 | 0.182 |

To evaluate the effect of the datasets distribution, the AdaFair model was retrained on a downsampled dataset and on a dataset augmented with generated counterfactuals. Table 3 reports the performance of the resulting models specifically on the positive non-protected instances. The results indicate that these modifications did not lead to consistent improvements across the models; however, substantial variability in performance was observed between individual instances. To examine whether these differences correspond to the position of instances in the feature space, their t-SNE embeddings are shown in Figure 2.

Table 3: Per-instance accuracy across models for the positive non-protected subgroup.

| | Instance | Original | Downsampled | Combined | Mean | Std |
|---|---|---|---|---|---|---|
| 0 | Inst3 | 1.0 | 1.0 | 1.0 | 1.000 | 0.000 |
| 1 | Inst6 | 0.4 | 0.2 | 0.0 | 0.200 | 0.200 |
| 2 | Inst5 | 0.6 | 0.0 | 0.0 | 0.200 | 0.346 |
| 3 | Inst7 | 0.6 | 0.4 | 0.8 | 0.600 | 0.200 |
| 4 | Inst2 | 0.6 | 0.4 | 0.4 | 0.467 | 0.115 |
| 5 | Inst1 | 0.2 | 0.4 | 0.0 | 0.200 | 0.200 |
| 6 | Inst4 | 0.4 | 0.4 | 0.0 | 0.267 | 0.231 |

These results indicate that instances located closer to the positive protected group exhibit higher predictive performance, whereas instances embedded within the majority distribution are the least accurately detected. Furthermore, Figure 3 demonstrates that the generated counterfactuals are positioned outside the majority distribution, suggesting that an instance's location within the majority-class regions is the primary factor contributing to its poor performance.

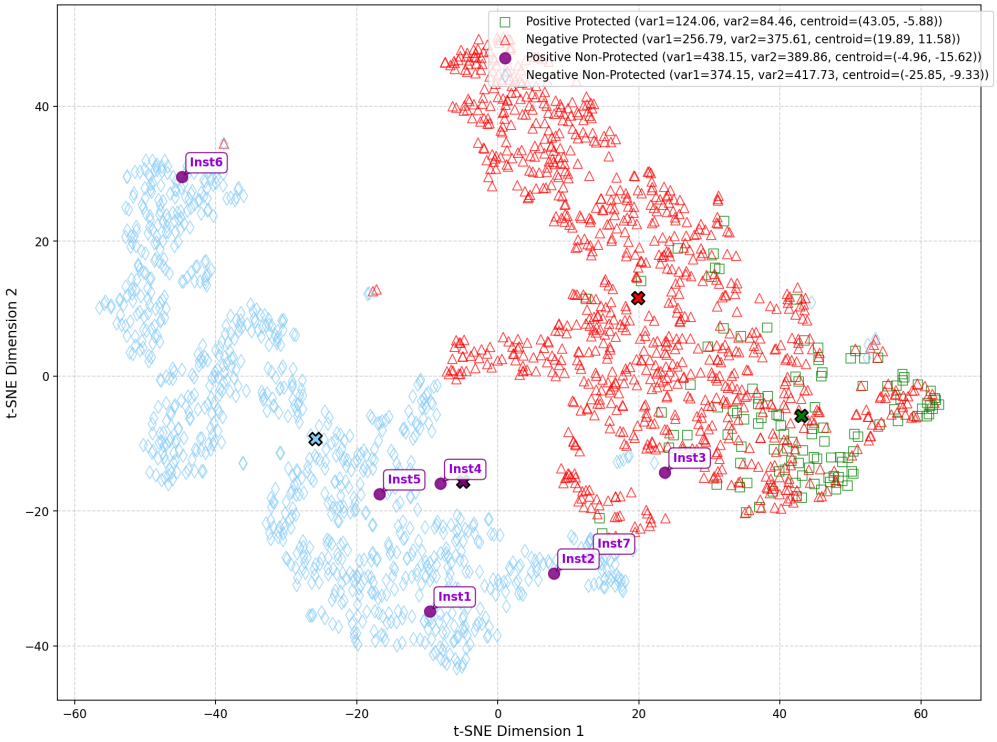


Figure 2: t-SNE visualization of the Communities Crime dataset including the `racepctblack` attribute. Instance tags are annotated for the positive non-protected group, and centroids and variances are shown for each subgroup.

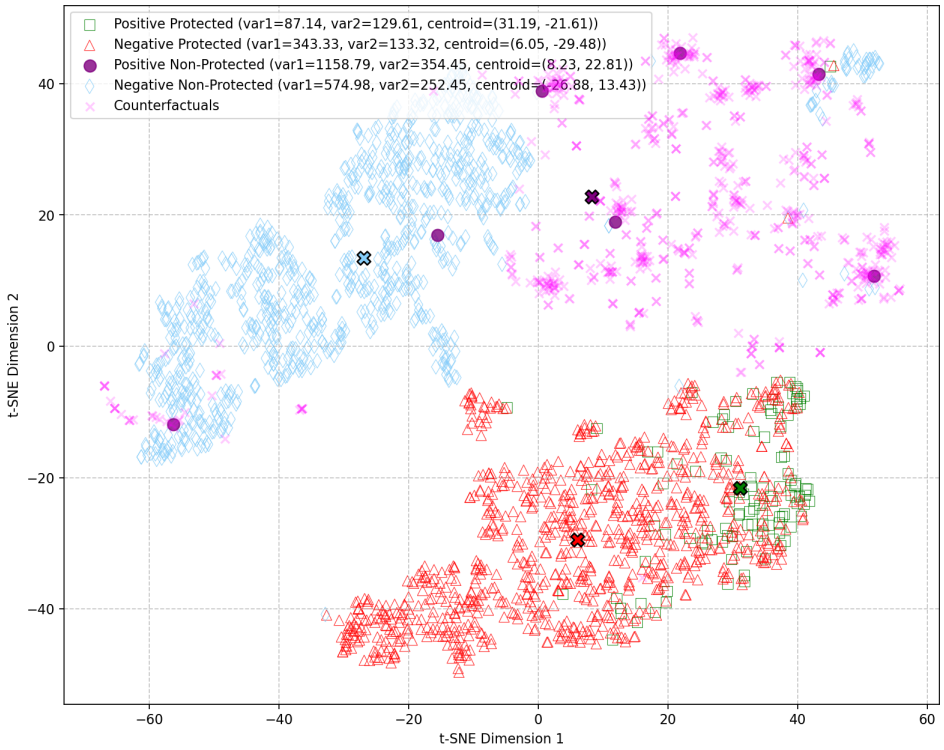


Figure 3: t-SNE plot of Communities and Crime dataset. Original instances are colored by subgroup (protected/non-protected, positive/negative). Counterfactuals are shown in magenta. Centroids and variances per group are annotated.

## 5 Discussion

This study investigated why AdaFair exhibits substantial performance degradation on the positive non-protected subpopulation in the Communities and Crime dataset. The counterfactual analysis revealed that `racepctblack` is the most important feature for flipping predictions when allowed to vary. When excluded, the model relies on proxy features strongly correlated with race (`PctKids2Par` and `PctIlleg`, $|r| \approx 0.50$). However, the most critical finding from our t-SNE analysis is that instance location within the feature space is the primary determinant of performance: positive non-protected instances embedded within the negative majority distribution achieve substantially lower accuracy than those near the positive protected group. Generated counterfactuals falling outside the majority distribution confirm that successful predictions require moving instances away from majority-class regions. Neither downsampling nor counterfactual augmentation yielded consistent improvements, indicating that performance is fundamentally constrained by unfavorable spatial distribution. These findings demonstrate that AdaFair struggles when extreme class imbalance combines with protected attribute imbalance, particularly when minority class instances are embedded within majority-class regions. However, certain limitations should be acknowledged. The reliance on t-SNE for visualization introduces potential distortion of the true high-dimensional relationships, as t-SNE preserves local but not necessarily global structure Wang et al. [2020]. Additionally, the counterfactual analysis depends on DiCE's optimization process and distance metrics, which may not fully capture all mechanisms by which AdaFair makes predictions. Future work should investigate whether preprocessing techniques that specifically address minority instances embedded in majority-class regions, such as FAWOS (Filtered-ADASYN with Oversampling) Napierała et al. [2010], can improve AdaFair's performance when applied prior to model training. Additionally, research should explore how such distributional problems can be identified and mitigated during the data collection process itself, potentially preventing these fairness issues before model training begins.

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